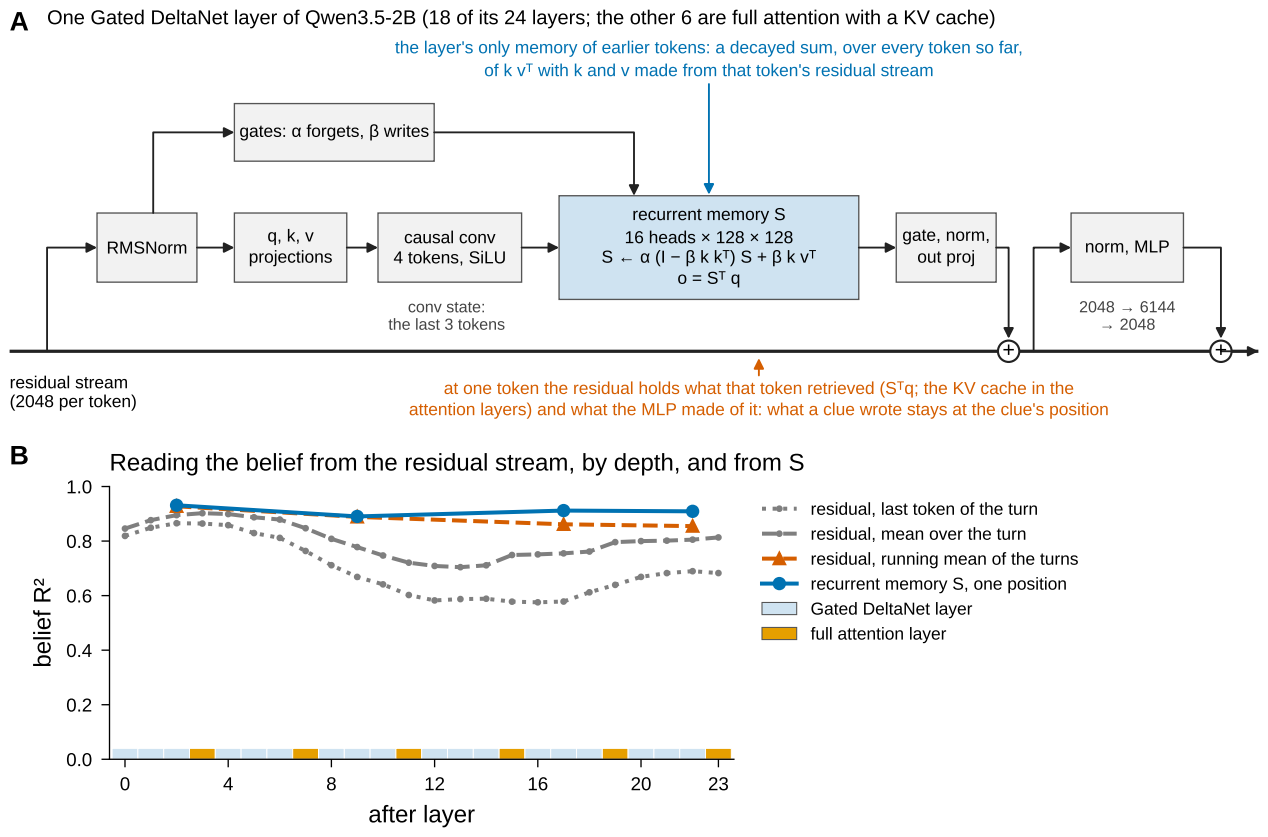
A
One Gated DeltaNet layer of Qwen3.5-2B (18 of its 24 layers; the other 6 are full attention with a KV cache)
the layer's only memory of earlier tokens: a decayed sum, over every token so far,
of k vᵀ with k and v made from that token's residual stream
gates: α forgets, β writes
RMSNorm
q, k, v
projections
causal conv
4 tokens, SiLU
recurrent memory S
16 heads × 128 × 128
S ← α (I − β k kᵀ) S + β k vᵀ
o = Sᵀ q
gate, norm,
out proj
norm, MLP
2048 → 6144
→ 2048
conv state:
the last 3 tokens
residual stream
(2048 per token)
at one token the residual holds what that token retrieved (SᵀqS; the KV cache in the
attention layers) and what the MLP made of it: what a clue wrote stays at the clue's position
B
Reading the belief from the residual stream, by depth, and from S
belief R²
1.0
0.8
0.6
0.4
0.2
0.0
0
4
8
12
16
20
23
after layer
residual, last token of the turn
residual, mean over the turn
residual, running mean of the turns
recurrent memory S, one position
Gated DeltaNet layer
full attention layer